Retail Business Performance & Profitability Analysis

Prepared by: Geetanjali Kaushik

Tools: Python | SQL | Power BI

Project Objectives

To develop an end-to-end retail analytics solution using Python for data preprocessing, SQL for profitability and trend analysis, and Power BI for interactive visualization, aimed at identifying profit-draining categories, optimizing inventory turnover, and understanding seasonal product behavior.

Dataset Overview

- **Retail Transactional Sales Dataset**

This dataset contains detailed transactional records from multiple shopping mall locations. Each row represents an individual customer purchase transaction, capturing sales, customer, and product-level information.

Correlation Analysis

```
# Correlation Analysis
correlation = df[['inventory_days','profit']].corr()
print (correlation)
```

```
                inventory_days    profit
inventory_days        1.000000 -0.010586
profit               -0.010586  1.000000
```

- **Correlation** shows the relationship between two variables.
- **+1** → Strong positive relationship
- **-1** → Strong negative relationship
- **0** → No relationship

Heatmap Visualization – Key Insights

Darker colors show **high sales/profit**, lighter colors show lower performance.

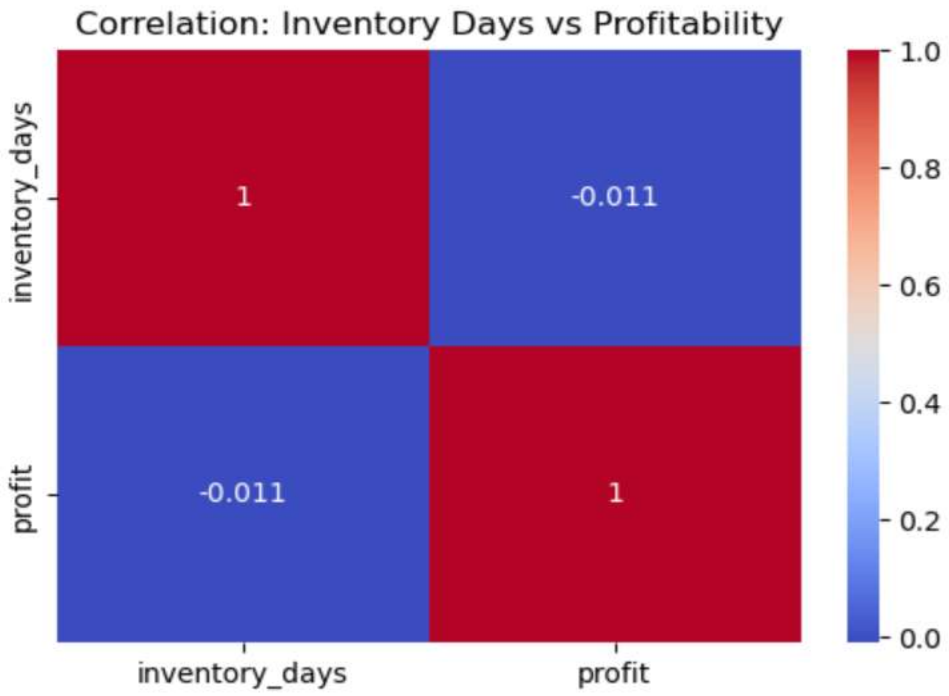
Clothing category appears consistently strong.
2023 shows lighter shades → overall sales decline.
Some categories have high sales but lower profit → margin improvement needed.
Certain products show longer inventory days → risk of overstock.

```python
# Heatmap Visualization (portfolio level)
plt.figure(figsize=(6,4))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation: Inventory Days vs Profitability")

plt.show()
```

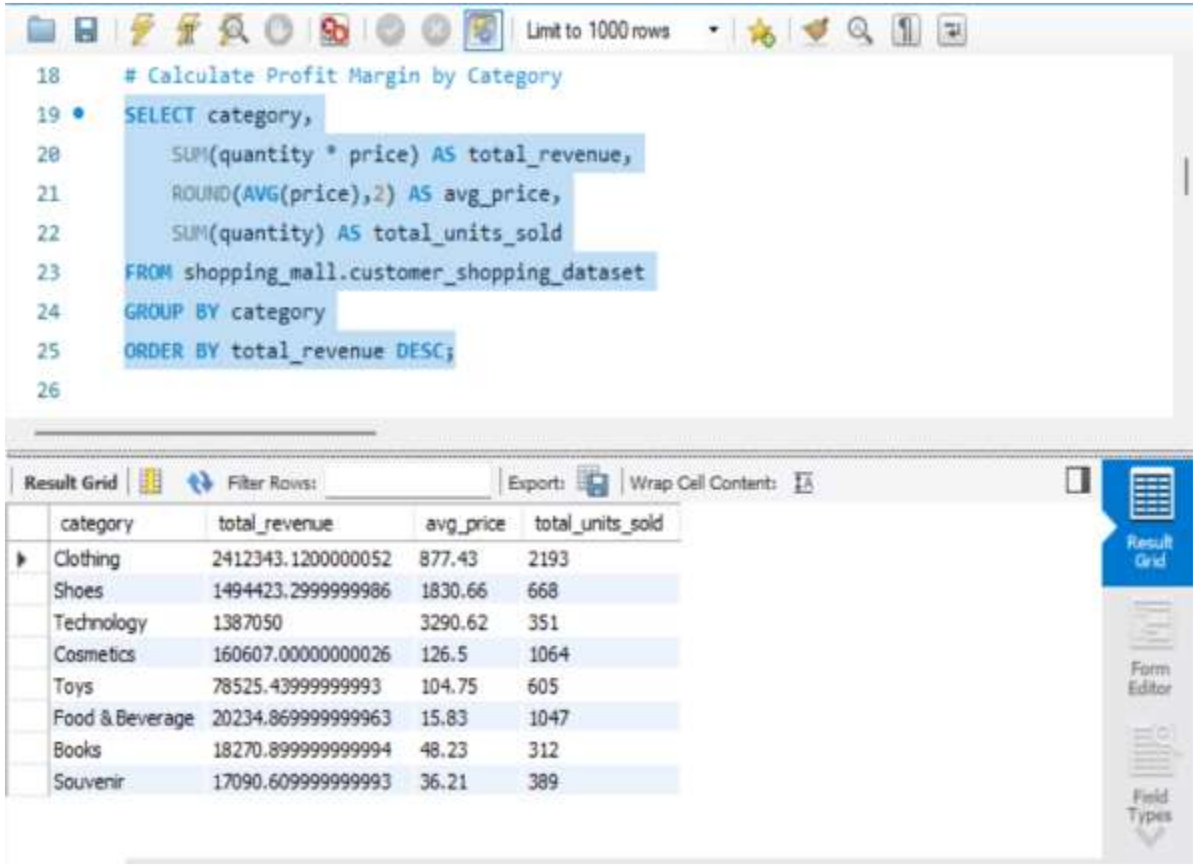


SQL Retail Sales & Customer Insights Analysis - Objectives

- ◆ To clean and validate raw transactional retail data
- ◆ To identify and handle missing (NULL) records
- ◆ To calculate revenue using Quantity × Price
- ◆ To analyze sales performance by product category
- ◆ To evaluate shopping mall-wise revenue contribution
- ◆ To identify top-performing categories using ranking functions
- ◆ To analyze monthly sales trends for seasonal insights
- ◆ To compute key business KPIs such as Average Bill Value
- ◆ To perform customer segmentation based on gender
- ◆ To transform raw data into actionable business insights using SQL

Profitability & Revenue Analysis (SQL)

- **Business Insights**:-
- Calculated total revenue using Quantity × Price.
- Identified top revenue-generating product categories.
- Estimated profit contribution by category (based on cost assumption).
- Measured revenue contribution percentage of each category.



Profit Margin Visualization

Profit Margin Insights:

By assuming a **30% cost structure**, the analysis highlights which product categories contribute the most to profitability and where businesses can improve pricing, inventory management, and cost efficiency to maximize overall revenue.

- **Formula Used:**
- Revenue = Quantity × Price
- Estimated Cost = 30% of Revenue
- Estimated Profit = 70% of Revenue
- Profit Margin % = 70%

```sql
# Profit Margin (Assuming of 30% Cost)
SELECT category,
    SUM(quantity * price) AS total_revenue,
    SUM(quantity * price) * 0.30 AS estimated_cost,
    SUM(quantity * price) * 0.70 AS estimated_profit,
    70 AS profit_margin_percent
FROM shopping_mall.customer_shopping_dataset
GROUP BY category
ORDER BY profit_margin_percent DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_revenue	estimated_cost	estimated_profit	profit_margin_percent
Clothing	2412343.1200000052	723702.9360000015	1688640.1840000036	70
Shoes	1494423.2999999986	448326.9899999996	1046096.309999999	70
Books	18270.899999999994	5481.269999999998	12789.629999999996	70
Cosmetics	160607.00000000026	48182.10000000008	112424.90000000018	70
Food & Beverage	20234.869999999963	6070.460999999988	14164.408999999972	70
Toys	78525.43999999993	23557.631999999998	54967.807999999946	70
Technology	1387050	416115	970934.999999999	70
Souvenir	17090.609999999993	5127.182999999998	11963.426999999994	70

Revenue Contribution % by Category

- **<u>Revenue Insights:-</u>**
- The top 5 categories contribute the largest share of total revenue, indicating strong customer demand and consistent purchasing behavior in these product segments.
- These high-performing categories play a critical role in overall business profitability and should be prioritized for inventory availability and promotional strategies.
- Focusing marketing efforts and stock management on these categories can significantly improve sales performance and customer satisfaction.
- The analysis helps businesses identify their core revenue drivers and allocate resources more effectively to maximize growth.

```sql
# Top 5 Revenue Generating Categories
SELECT category,
    SUM(quantity * price) AS total_revenue
FROM shopping_mall.customer_shopping_dataset
GROUP BY category
ORDER BY total_revenue DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

category	total_revenue
Clothing	2412343.1200000052
Shoes	1494423.2999999986
Technology	1387050
Cosmetics	160607.00000000026
Toys	78525.43999999993

Revenue by Shopping Mall Location

- **Insights:-**
- Top-performing malls generate the highest revenue, indicating strong customer traffic and purchasing activity.
- Mall-wise revenue comparison helps identify high-demand locations for better expansion and store planning.
- High units sold but lower revenue may indicate lower pricing or heavy discounts.
- Low-performing malls may need targeted promotions or marketing strategies.
- Revenue insights help optimize inventory distribution across locations.

```
# Revenue by Shopping Mall Location
SELECT shopping_mall,
    SUM(quantity * price) AS total_revenue,
    SUM(quantity) AS total_units_sold
FROM shopping_mall.customer_shopping_dataset
GROUP BY shopping_mall
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

shopping_mall	total_revenue	total_units_sold
Mall of Istanbul	1260911.4999999995	1342
Kanyon	1131274.02	1433
Metrocity	715906.4999999995	1035
Metropol AVM	497218.6799999999	645
Istinye Park	457009.2100000001	572
Zorlu Center	329372.4799999998	311
Forum Istanbul	318948.8000000001	330
Emaar Square Mall	318542.92	351
Cevahir AVM	281015.55000000005	295
Viaport Outlet	278345.58000000013	315

Result 11

Mall-wise Category Performance

Insights:

- Category demand varies across malls, showing different customer preferences by location.
- Top categories in each mall generate the highest revenue, driving location-wise sales.
- Some categories perform well only in specific malls, suggesting the need for location-based product strategies.
- Low-performing categories may require promotions or inventory adjustments.

```sql
# Mall-wise Category Performance
SELECT shopping_mall,category,
    SUM(quantity * price) AS total_revenue
FROM
    shopping_mall.customer_shopping_dataset
GROUP BY shopping_mall , category
ORDER BY shopping_mall , total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

shopping_mall	category	total_revenue
Cevahir AVM	Technology	143850
Cevahir AVM	Clothing	99026.40000000004
Cevahir AVM	Shoes	24606.969999999998
Cevahir AVM	Cosmetics	8823.219999999998
Cevahir AVM	Toys	3368.960000000001
Cevahir AVM	Food & Beverage	826.34
Cevahir AVM	Souvenir	316.71
Cevahir AVM	Books	196.95000000000002
Emaar Square Mall	Clothing	136536.4
Emaar Square Mall	Shoes	103829.40999999999

Gender-wise Revenue Analysis

- **Insights:**
- Gender-wise revenue shows which customer group contributes more to total sales.
- Comparing revenue and customer count reveals spending behavior.
- Higher revenue from one gender may indicate stronger product preference.
- Helps businesses create targeted marketing strategies.

```sql
# Gender-wise Revenue Analysis
SELECT
    gender,
    SUM(quantity * price) AS total_revenue,
    COUNT(DISTINCT customer_id) AS total_customers
FROM shopping_mall.customer_shopping_dataset
GROUP BY gender;
```

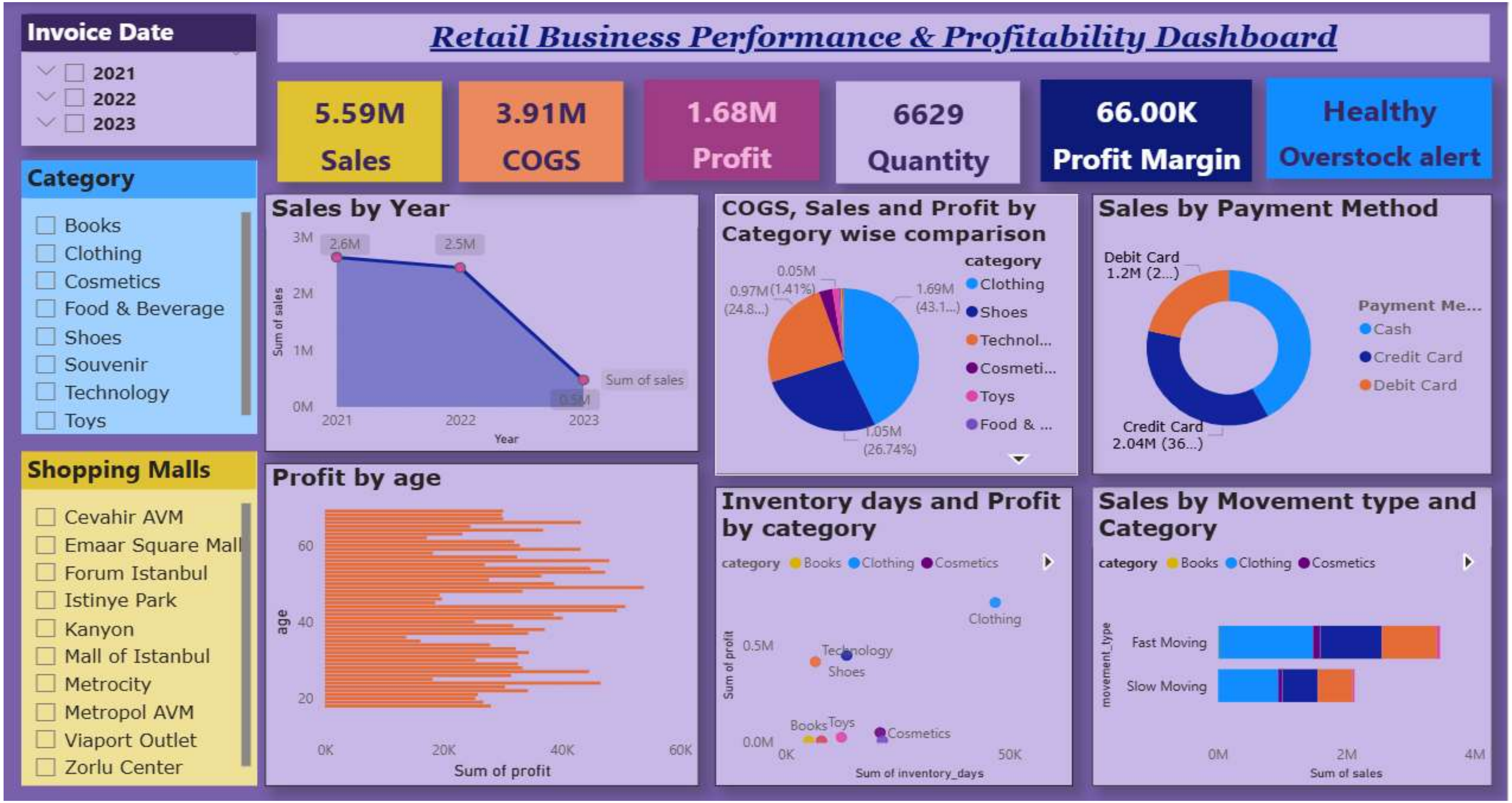
Result Grid Filter Rows: Export: Wrap Cell Content:

gender	total_revenue	total_customers
Female	3277233.6999999974	1339
Male	2311311.54	861

Retail Business Performance & Profitability Dashboard – Overview

- This dashboard analyzes retail sales, profit, inventory performance, and customer payment behavior. It highlights key KPIs like **Sales, COGS, Profit, Quantity, and Profit Margin** to show overall business performance.
- The visuals help identify **sales trends over time, profitable product categories, payment preferences, and inventory risks**. It also compares **inventory days with profitability** to detect potential overstock issues.
- Overall, the dashboard helps businesses **monitor performance, optimize inventory, and make better data-driven decisions.**

Power BI Retail Business Analysis Dashboard



Retail Business Strategies

- **Invest in top-performing categories** like Clothing to maximize sales and profit.
- **Improve inventory turnover** by reducing stock of slow-moving products.
- **Boost sales through promotions** and marketing campaigns, especially after the 2023 decline.
- **Target the 30–50 age group** with personalized offers and loyalty programs.
- **Ensure availability of fast-moving products** to maintain steady revenue.
- **Strengthen digital payment options** to improve customer convenience and transaction speed.
- **Analyze mall-wise performance** to identify high-performing locations and optimize store strategies.

KPI driving Retail Performance

Key Insights:-

Total Sales: 5.59M showing strong overall revenue performance.

COGS: 3.91M indicating the total cost involved in generating sales.

Total Profit: 1.68M reflecting healthy profitability.

Total Quantity Sold: 6629 units across all product categories.

Profit Margin: 66K indicating strong margin performance.

Overstock Alert: Current inventory status is **Healthy**, showing no major overstock risk.

Explore Retail Insights for attractive slicers

- **The dashboard includes slicers for:-**

✓Invoice date

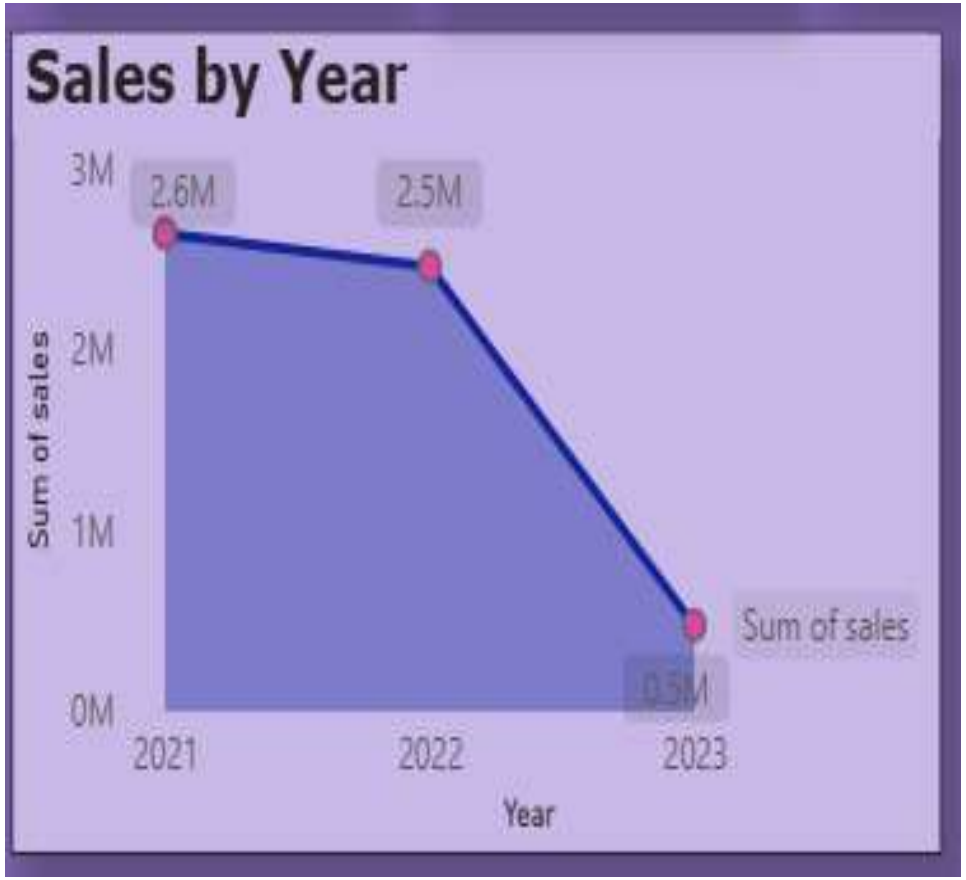
✓Category

✓Shopping malls

- These interactive filters allow users to **analyze retail performance dynamically by year, product category, and mall location**, helping businesses make **better inventory, sales, and expansion decisions**.

Sales Trend Analysis

- **<u>Insights:-</u>**
- **Sales were highest in 2021** at around **2.6M**, showing strong retail performance.
- **2022 sales slightly declined** to **2.5M**, indicating a small slowdown but still stable performance.
- **2023 shows a sharp drop** to around **0.5M**, suggesting a significant decrease in sales.
- The trend indicates **business performance weakened significantly in 2023**, which may require further investigation into demand, inventory, or market conditions.

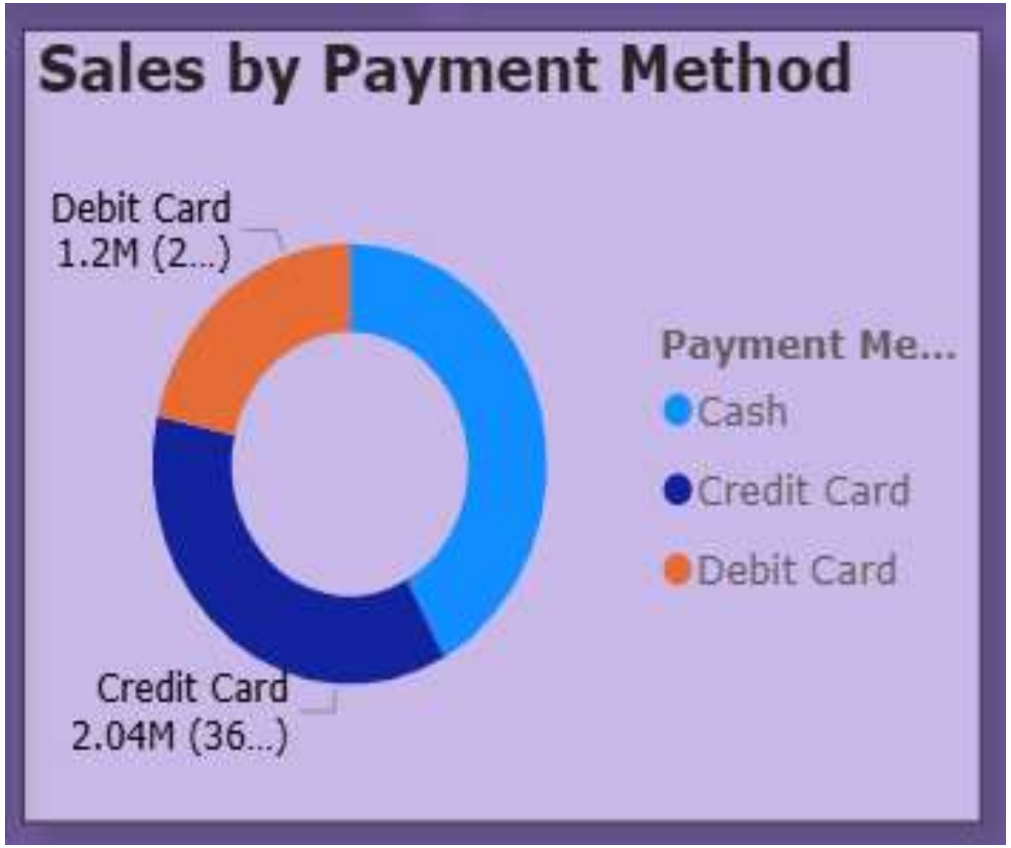


Category-wise Sales, COGS & Profit Comparison

- **Insights:-**
- **Clothing dominates the sales** with about **1.69M (43%)**, making it the top-performing category.
- **Shoes contribute the second-highest share** with **1.05M (26.7%)**, showing strong customer demand
- **Technology accounts for around 0.97M (24.8%)**, indicating steady performance.
- **Cosmetics, Toys, and Food & Beverage contribute very small shares**, suggesting lower demand or limited sales impact.

Sales Distribution by Payment Method

- **Insights:-**
- **Cash is the most preferred payment method**, contributing the largest share of total sales.
- **Credit Card payments account for around 2.04M**, showing strong usage of card-based transactions.
- **Debit Card contributes the smallest share (~1.2M)** compared to other payment methods.
- The data suggests that customers still rely heavily on **cash and credit card transactions** in retail purchases.



Profit Distribution by Customer Age

- **Insights:-**

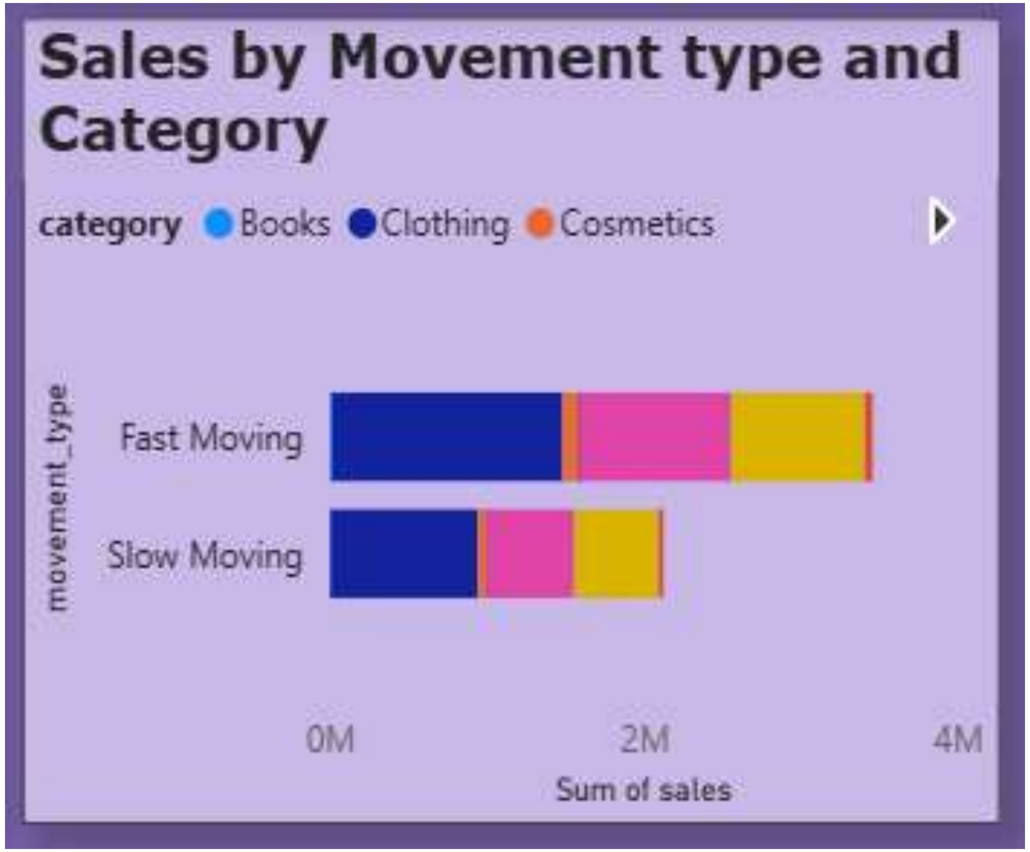
- Profit is generated across a wide range of age groups, showing diverse customer participation.
- Customers between **30–50 years** appear to contribute higher profit compared to younger age groups.
- Middle-aged customers tend to have **strong purchasing power and consistent spending behavior**.
- Younger customers contribute comparatively lower profit levels.

Inventory Days vs Profit by Category

- **<u>Insights:-</u>**
- **Clothing generates the highest profit**, but it also has the **highest inventory holding days**, indicating strong demand despite longer stock periods.
- **Technology and Shoes show moderate profit with moderate inventory days**, suggesting balanced stock movement.
- **Books, Toys, and Cosmetics generate very low profit**, even with lower inventory days.
- Categories with **higher inventory days but low profit may indicate slow-moving products**.

Sales Performance by Movement Type and Category

- **<u>Insights:</u>-**
- **Fast-moving products generate higher sales** compared to slow-moving items.
- **Clothing contributes the largest share of sales** among both movement types.
- **Slow-moving products still contribute to revenue**, but their sales volume is significantly lower.
- Categories with slow movement may require **better inventory management or promotional strategies**.



Retail Business Recommendation

- Focus on **high-performing categories like Clothing** to increase revenue and profit.
- Improve **inventory management** by reducing slow-moving products.
- Investigate the **sales decline in 2023** and introduce promotional strategies.
- Target **customers aged 30–50**, as they contribute higher profits.
- Promote **fast-moving products** and maintain adequate stock levels.

Thank You ☺